EE2015: Electric Circuits and Networks

Tutorial 8 Answer Key
(October 17, 2025)

1. $v_C(t) = 4.8\,e^{-t} - 3.5\,e^{-2t} - 0.3\cos 2t - 0.1\sin 2t$ for $t \geq 0$
2. $\sin(100t)\,\mathrm{V}$
3.
4. $Z_{\text{in}} = 3.22 - j11.07\,\Omega$
5. $\mathbf{V}_{\text{Th}} = 55\angle -90^0\,\mathrm{V}$, $\mathbf{Z}_{\text{Th}} = 4 - j\frac{2}{3}\,\Omega$
6. $v_C(t) = 7.809\,\cos(2t - 38.66^o)\,\mathrm{V}$, $i_L(t) = 1.5617\,\cos(2t - 38.66^o)\,\mathrm{A}$
7.